Genevieve A. Mortensen

Curriculum Vitae | Bioinformatics · Computational Omics · Biomedical AI
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PROFILE

Ph.D. candidate in Bioinformatics and M.S. candidate in Computer Science at Indiana University, bridging machine learning, multi-omics, and clinical data. Builds reproducible Python/R pipelines on HPC/GPU systems for longitudinal ICU modeling, metagenomics, metabolomics, genomics, and biomedical language models. Brings two Zoetis omics bioinformatics internships and three years of cGMP and CLIA/CAP/ISO-regulated molecular laboratory experience.

Research interests: physiological representation learning; longitudinal and survival modeling; electronic health records; privacy-preserving biomedical AI; metagenomics and microbiome; comparative metabolomics of aging; genomics and synthetic biology; reproducible scientific software.

EDUCATION

Ph.D. in Informatics, Bioinformatics Track — Expected May 2027
Indiana University Bloomington, Luddy School of Informatics, Computing, and Engineering | Advisor: Haixu Tang
Dissertation: Hierarchical physiological representation learning and survival modeling from longitudinal intensive-care data.

M.S. in Computer Science — Expected Dec 2026
Indiana University Bloomington

B.S. in Cell and Molecular Biology — Dec 2020
The University of Texas at Austin | Research advisor: Jeffrey E. Barrick

RESEARCH EXPERIENCE

Research Assistant, Tang Lab — Aug 2022 – Present
Indiana University Bloomington, Luddy School of Informatics, Computing, and Engineering | PI: Haixu Tang

- **Virtual patient (dissertation).** Design and train a hierarchical variational autoencoder for longitudinal MIMIC-IV ICU data, with physiological-subsystem encoders, system- and patient-level latent variables, mirrored decoders, and a Cox/DeepSurv survival head.
- **Clinical data engineering.** Built hourly extraction and clinically informed preprocessing (values, observation masks, time-since-measured features) with streaming GPU training on HPC, ablation studies, calibration, and fixed-horizon AUC/Brier evaluation.
- **Gestational disease microbiome.** Developed MGPipe, a reproducible shotgun-metagenomics workflow, and built predictive models of maternal gut-microbiome signatures (first-author, *Microbiology Spectrum* 2026).
- **Biomedical language models.** Co-developed ADetectoLocum, a locally deployable LLM pipeline for voice-based Alzheimer's risk detection, and evaluated quality and bias across 50,000+ MIMIC-IV discharge summaries (AMIA 2025; IEEE BHI 2025).
- **Comparative metabolomics of aging.** Lead cross-species analysis of Drosophila, mouse, and elephant aging using age-effect modeling, trajectory analysis, pathway scoring, and concordance testing (bioRxiv 2026).
- **Collaborative clinical AI.** Contributed to personalized real-time hemodynamic targets in critical care (*Nature Communications* 2025) and an LLM benchmark for analytical-method lifecycle management (ACL Findings 2026).

Pharmacogenomics Research Intern May 2023 – Aug 2023

Indiana University School of Medicine, Department of Clinical Pharmacology, Indianapolis, IN | Investigators: Nicholas Powell, Todd Skaar

- Developed miRilizer, a web-hosted R Shiny application and pipeline for mRNA–miRNA interactome analysis in human hepatocytes, supporting target discovery and pharmacogenomics research.
- Presented the application and results to clinical pharmacology researchers and reported progress at lab meetings.

Research Assistant, Barrick Lab Jan 2020 – Jan 2021

The University of Texas at Austin | PI: Jeffrey E. Barrick

- Adapted bacterial genome-resequencing workflows and analyzed NGS data on Texas Advanced Computing Center resources; modeled transcription and population growth in C++, Python, and R.
- Built an interactive web simulator for culture saturation and cellular-load maximization, plus Jupyter and R Shiny tools to share model behavior with collaborators.

iGEM Undergraduate Researcher Jan 2019 – Jun 2020

The University of Texas at Austin | Advisors: Jeffrey E. Barrick, Dennis M. Mishler | Funding: NSF BEACON

- Analyzed E. coli growth and gene-expression data to measure the evolutionary stability and cellular burden of engineered BioBrick constructs (R, Python, R Shiny).
- Earned equal-contribution authorship on the resulting *Nature Communications* (2024) article; team won a Gold Medal and Best Measurement prize at the iGEM Giant Jamboree.

Laboratory Technician, Barrick Lab May 2019 – Sep 2020

The University of Texas at Austin

- Supported BSL-2 microbiology research: bacterial culture, transformations, nucleic-acid extraction, media preparation, automated-equipment optimization, and inventory control.

Summer Undergraduate Researcher, Wilke Lab Jun 2018 – Sep 2018

The University of Texas at Austin | Mentor: Ashley Teufel | PI: Claus O. Wilke

- Modeled Lotka–Volterra systems and eigenvalue-based evolutionary-state dynamics in Python and R, and visualized simulated trajectories to examine stability and state transitions.

ONGOING PROJECTS AND COLLABORATIONS

Mitochondrial Genomic Variation and Cardiovascular Comorbidities Ongoing

With Indiana University School of Medicine

- Investigating associations between cardiovascular comorbidities and somatic mitochondrial copy-number and genomic variation using All of Us genomic and clinical data; poster at the 2025 NHGRI CEGS Meeting.

AI-Guided Mass Spectrometry Instrument Repair Ongoing

With Agilent Technologies

- Developing a machine-learning tool that predicts likely replacement components to guide diagnosis and repair of mass spectrometry instruments.

INDUSTRY AND CLINICAL LABORATORY EXPERIENCE

Omics Bioinformatics Data Analyst Intern (2 summers) May–Aug 2024; May–Aug 2025

Zoetis, Systems Biology / Omics Bioinformatics, Kalamazoo, MI

- **2025:** Built an end-to-end metabolomics pipeline for reproducible processing, statistics, visualization, and reporting; developed the front-end reporting module for sequence-homology detection.
- **2025:** Supported antigen and vaccine-target discovery with reverse vaccinology and mRNA–miRNA interactome analysis, and created statistical metrics to evaluate the target-prediction pipeline.
- **2024:** Developed multi-omics reporting infrastructure in Python and R, and provided Bash/Python back-end support for NGS, microbiome, and reference-literature pipelines through regular design reviews.

Research Associate Sep 2021 – Aug 2022

Asuragen, a Bio-Techne company, Austin, TX

- Developed and quality-controlled custom molecular controls for infectious and genetic disease assays (qPCR, RNA-seq, DNA sequencing) under cGMP in a CLIA/CAP-accredited laboratory; validated new products.
- Built Python, Shell, and VBA tools on HPC for proprietary-sequence screening, secure data management, inventory tracking, and customer-audit readiness.

Manufacturing Scientist Apr 2021 – Sep 2021

Agilent Technologies, Cedar Creek, TX

- Manufactured and quality-controlled IVD/RUO oncology reagents under cGMP; analyzed FISH, qPCR, endpoint-PCR, and mass spectrometry QC data and built VBA tools for reagent-expiration tracking.

Molecular Technologist May 2020 – Apr 2021

Sonic Reference Laboratory, Austin, TX

- Performed high-complexity molecular diagnostic testing (qPCR, endpoint PCR, RNA workflows) in a CAP/ISO-accredited clinical laboratory during high-throughput COVID-19 testing.
- Automated multiplex qPCR threshold calling in R and Python, and built secure instrument-to-database transfer with VBA/PostgreSQL for liquid-handling robotics.

PUBLICATIONS

** Equal contribution. Full record on Google Scholar.*

Peer-reviewed journal articles

1. **Mortensen, G. A.**, Schmidt, H., Radivojac, P., Ye, Y., & Haas, D. M. (2026). Metagenomic profiling and predictive modeling of the gut microbiome reveal signatures of gestational disease. *Microbiology Spectrum*, 14, e03155-25. doi:10.1128/spectrum.03155-25
2. Sun, Y., Li, J., Liu, X., **Mortensen, G. A.**, Gu, X., Adams, D. C., Tang, H., Su, J., Liu, Z., Sun, D., & Meng, L. (2025). The HM-TARGET personalised real-time haemodynamic targets in critical care. *Nature Communications*, 16, 7307. doi:10.1038/s41467-025-62527-x
3. Radde, N.*, **Mortensen, G. A.***, Bhat, D., Shah, S., Clements, J. J., Leonard, S. P., McGuffie, M. J., Mishler, D. M., & Barrick, J. E. (2024). Measuring the burden of hundreds of BioBricks defines an evolutionary limit on constructability in synthetic biology. *Nature Communications*, 15, 6242. doi:10.1038/s41467-024-50639-9

Peer-reviewed conference proceedings

4. Chen, X., Monshizadeh, M., Zhang, C., Lang, J., Wu, Y., **Mortensen, G.**, Liu, X., & Tang, H. (2026). LLMs as Lab Engineers: A benchmark for analytical method lifecycle management. *Findings of the Association for Computational Linguistics: ACL 2026*, 40533–40553. doi:10.18653/v1/2026.findings-acl.2015
5. **Mortensen, G. A.**, & Zhu, R. (2025). Early Alzheimer's detection through voice analysis: Harnessing locally deployable LLMs via ADetectoLocum, a privacy-preserving diagnostic system. *AMIA Joint Summits on Translational Science Proceedings 2025*, 365–374. PMID 40502222. PMC12150716
6. Zhu, R., **Mortensen, G.**, Fan, Y., & Tang, H. (2025). Illuminating the unseen: A large-scale exploration of bias in ICU discharge summaries via language models. *2025 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, 1–8. doi:10.1109/BHI67747.2025.11269458

Preprints

7. **Mortensen, G. A.**, Montgomery, E. E., Ng'Ombwa, I., Smoot, S. R., Stephenson, D., Kaufman, T. C., Nemkov, T., D'Alessandro, A., Hurley, L. M., Tennessen, J. M., Tang, H., & Chusyd, D. E. (2026). Comparative metabolomics identifies recurrent age-associated pathway remodeling across species. *bioRxiv*. doi:10.64898/2026.06.23.733839

OPEN-SOURCE SOFTWARE

- MGPipe — Reproducible shotgun-metagenomics workflow: read QC, host-read removal, taxonomic and functional profiling, analysis-ready outputs.
- ADetectoLocum — Locally deployable, privacy-preserving LLM and machine-learning pipeline for voice-based Alzheimer's risk detection.
- Aging_Metabolome — Cross-species metabolomics workflow for age-effect modeling, trajectory analysis, pathway scoring, and visualization.
- miRilizer — R Shiny application for interactive exploration of mRNA–miRNA interactions in human hepatocytes.
- iGEM 2019 Burden Analysis Software — Archived code and interactive burden models supporting the BioBricks constructability study.
- genome-assembly and sc-seq — Python workflows for genome assembly and annotation, and notebooks for machine-learning analysis of single-cell RNA/ATAC-seq data.

PRESENTATIONS

Oral presentations

2026 Lightning talk and poster. Center for Bioinformatics Research Retreat, Indiana University, Bloomington, IN.

2025 "Early Alzheimer's Detection Through Voice Analysis: Harnessing Locally Deployable LLMs via ADetectoLocum." AMIA 2025 Informatics Summit, Pittsburgh, PA.

2020 Evolutionary stability and burden of engineered genetic constructs. NSF BEACON Congress; UT Austin Undergraduate Research Symposium, Austin, TX.

2019 iGEM team presentation on BioBrick construct stability and burden. iGEM Giant Jamboree, Boston, MA; UT Austin Synthetic Biology Conference, Austin, TX.

Posters

2026 Luddy ICE 25th Annual Celebration, Indiana University, Bloomington, IN.

2025 "Investigating the Impact of Cardiovascular Comorbidities on Mitochondrial Genomic Variation in Somatic Cells." NHGRI Centers of Excellence in Genomic Science (CEGS) Annual Meeting, Pittsburgh, PA.

2025 Center for Bioinformatics Research Retreat, Indiana University, Bloomington, IN.

2024 AMIA 2024 Annual Symposium, San Francisco, CA.

TEACHING AND MENTORING

Associate Instructor, Introductory Python Programming Aug 2022 – May 2023
Indiana University Bloomington

- Independently taught course sections, graded and mentored students in programming and biomedical data analysis, and coordinated a team of four peer instructors.

Research Instructor, Laboratory Experience in Molecular Biology Jan 2021 – May 2021
Texas Institute for Discovery Education in Science (TIDES), UT Austin | Instructor: Mona Mehdy

- Held office hours and tutorials guiding upper-division students through biological data analysis in a research-based plant molecular biology course.

Teaching Facilitator, Cell Biology May 2019 – Sep 2020
Peer-Led Undergraduate Study, UT Austin | Instructor: Clarence Chan

- Designed weekly study guides and led peer-learning sessions, adapting review activities with the course instructor.

Research mentees

Sheena Williams, undergraduate (Cybersecurity), IU — biomedical-record visualization, HIPAA-aligned data handling, secure computing. Fall 2023 – Spring 2024

Myat Aung, undergraduate (Informatics), IU IUUR STEM Program — AI for disease diagnosis. Summer 2023

HONORS AND AWARDS

Gold Medal, International Genetically Engineered Machine (iGEM) Competition — team award 2019

Best Measurement Special Prize, iGEM Competition — team award 2019

Nominee, Best Foundational Advancement, iGEM Competition 2019

Undergraduate Research Travel Award, The University of Texas at Austin 2019

LEADERSHIP AND PROFESSIONAL SERVICE

Seminar Series Committee Coordinator Jan 2023 – Present

Center for Bioinformatics Research, *Indiana University*

- Coordinate invited seminars, speaker logistics, and communications; organized the 2025 and 2026 CBR research retreats.

President Aug 2023 – Dec 2025

Indiana University Bioinformatics Club

- Led seminars, technical workshops, and student research talks in partnership with the Center for Bioinformatics Research.

Technical Program Committee Member 2024

Workshop on AI Systems with Confidential Computing (AISCC), *NDSS Symposium*

Peer review

Journal: *IEEE Transactions on Computational Biology and Bioinformatics* (2025, 2026)

Conference: Pacific Symposium on Biocomputing (2024, 2025); IEEE EMBC (2026); International Conference on Intelligent Biology and Medicine (2026)

Professional affiliations

Women in Computing, Indiana University; Computer Science Club, Indiana University

TECHNICAL SKILLS

Programming: Python, R, SQL, Bash/Shell, C++, JavaScript, HTML, VBA, LaTeX

Machine learning and statistics: PyTorch, TensorFlow, scikit-learn, statsmodels, XGBoost; variational autoencoders, LLMs, representation learning, survival analysis (Cox/DeepSurv), mixed-effects models, calibration, clustering, dimensionality reduction, hypothesis testing

Biomedical and omics data: MIMIC-IV and EHR, longitudinal clinical data, metagenomics and microbiome, metabolomics, genomics, NGS, transcriptomics, pharmacogenomics, single-cell sequencing, synthetic biology

Platforms: Linux, HPC, GPU/CUDA, Git, Jupyter, R Shiny, PostgreSQL, pandas, NumPy, matplotlib, ggplot2, reproducible reporting

Laboratory and quality: qPCR, endpoint PCR, RNA/DNA sequencing, FISH, bacterial culture and transformation, nucleic-acid extraction, protein purification, cell culture, BSL-2, cGMP, CLIA/CAP/ISO practices

Instrumentation: Tecan M200 PRO; Applied Biosystems 7500 Fast Dx and QuantStudio 6/7 Flex; Roche MagNA Pure 96; Bio-Rad CFX96; Hamilton Microlab STAR; KingFisher Flex; Beckman Coulter Biomek i5

Communication: Adobe Illustrator, Microsoft Office

ADDITIONAL EXPERIENCE

Private Consultant Jun 2022 – Jan 2023

Goldfish Society of America

- Advised on chemical treatment protocols, morphology and longevity, habitat design, and algae culture; contributed to hobbyist education and outreach.

Mental Health Technician and Shift Lead Jan 2018 – May 2020

Recovery Unplugged, Austin, TX

- Supported behavioral-health and substance-use treatment through documentation, vital-sign monitoring, and medication-support protocols; led shifts of up to five staff and prepared for audits.